

Development of soil mineral classification using ontology mining

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ABSTRACT ORIGINAL

To maintain the quality of agriculture, soil minerals are playing an important role, and it is considered an unavoidable factor for crop growth. Soil is an organic matter together with mineral particles that help to regulate water, air, and other nutrients to the plants. For growing a quality crop, the composition of minerals needs to be appropriate to provide the sufficient level of nutrients. The composition of minerals is varied depending upon the types of crop being grown. Usually the knowledge about the composition is known to the experienced farmers and elders who try to designate the details to the followers. However, there is no guarantee for maximum level of knowledge transfer. Thus there arrives a demand for capturing these valuable knowledge for the agricultural novice. Due to the advent of new technology, capturing and storing these knowledge becomes easy and effective. Thus an attempt is made to capture the mineral content and its composition of soil to make it available as a single knowledge base. The designed knowledge base is expected to be available for novice farmers to guide and help them in the various agricultural processes and decision-making. In this paper, an ontology-based knowledge base is created for storing the details of soil composition with different minerals. The ontology provides a structured formalized knowledge which is mined for different patterns that subsequently concludes with recommendation of crops and the soil composition suitable for crops. A mining methodology is derived to fulfill the abovementioned objective. © 2021, Saudi Society for Geosciences.